

PCBs and enclose in suitable cabinets. As shown in the block diagram, place microphone MIC1 and loudspeaker LSP1 in the first room, and microphone MIC2, loudspeaker LSP2 and the entire electronics block plus the power supply in the second room. Keep the interconnection cables as short as possible—preferably shorter than 10 metres. These should be kept away from the cables of the mains power supply and other sources of

electromagnetic interference. Use a shielded cable for connecting the microphone to the circuit.

It is important that there is no acoustical feedback between the microphone and the loudspeaker on both the sides. So the microphones and the loudspeakers should not face each other. As the gain of each channel is not very high, the probability of acoustical feedback is low. In case of acoustical feedback, lower the

volume of the amplifier and change the positions of the microphones and loudspeakers.

The preferred power supply for the circuit is 6V or above but the circuit also works with regulated 5V from IC 7805 (not shown in the diagram). If a higher dynamic range is needed, the power supply should be 9V or even 12V. For power supply, you can use an AC-DC wall adaptor, dry batteries or rechargeable batteries. ●

Simple Water Temperature Indicator

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When bathing a baby, it's important that the water is neither too cold nor too hot. You test the water by dipping your fingers in the bath tub before putting the baby into it. Here is a simple circuit that indicates the water temperature, helping you to use water that is just right for your baby. Limit points for high and low temperatures can be set to any level. So this circuit can also be used to ensure optimal temperature for a room, greenhouse, pool, geyser, fish-tank or freezer.

Fig. 1 shows the circuit of the water temperature indicator. The circuit is built around dual op-amplifier IC TL072 (IC1), which is configured in comparator mode. The sensor used in the circuit is a thermistor—a two-pin device whose resistance varies with temperature. Thermistor TH1 has a resistance of 47 kilo-ohms at room temperature (25°C). Resistor R1 of the same value as TH1 is connected in series with TH1.

At 25°C, the resistance of

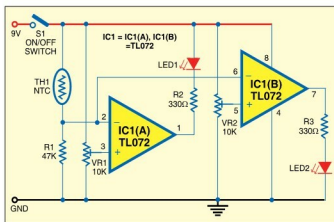


Fig. 1: Circuit of the water temperature indicator

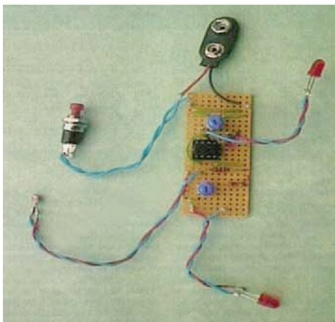


Fig. 2: Author's prototype

the thermistor is around 47 kilo-ohms. So the voltage at pin 2 of the first op-amplifier IC1(A) is approximately half the supply voltage. As the water temperature increases, the resistance of the thermistor decreases, while the resistance of R1 remains unchanged. As a result, the voltage at input pin 2 of IC1 increases. Conversely, a fall in the temperature of TH1 causes the voltage to fall at pin 2 of IC1.

The reference voltage at pin 3 of IC1 is set by using preset VR1. It is set such that at room temperature, the voltage at pin 2 (inverting input) is lower than the voltage at pin 3 (non-inverting input). As a result, the output at pin 1 is high and LED1 doesn't glow.

IC1(B) is the second op-amp, with pin 6 as the input. The reference voltage for the upper limit of water temperature at its pin 5 is set using preset VR2. It is set such that when the temperature is below the upper limit, the voltage at pin 5 is higher than the voltage at pin 6 and LED2 glows. But when the

